UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/345,217	06/30/2023	Si LIN	00333.0186.00US	6234

187287 7590 08/28/2025

Anova Law Group
21495 Ridgetop Circle
Suite 300
STERLING, VA 20166

EXAMINER
KOLB, NATHANIEL J

ART UNIT	PAPER NUMBER
2896	

NOTIFICATION DATE	DELIVERY MODE
08/28/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspto.anova@gmail.com
uspto@anovalaw.com

Office Action Summary	Application No. 18/345,217	Applicant(s) LIN et al.	
	Examiner NATHANIEL J KOLB	Art Unit 2896	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 02 Oct 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-12 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-12 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 30 June 2023 is/are: a)☐ accepted or b)☑ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Summary

2. Claims 1-12 are pending. Claims 1-12 are rejected herein. This is a First Action on the Merits.

Drawings

3. FIG. 2 is objected to for containing only unlabeled boxes (or other shapes).

Regarding the content of drawings 37 C.F.R. 1.83 (a) states:

> (a) The drawing in a nonprovisional application must show every feature of the invention specified in the claims. However, conventional features disclosed in the description and claims, where their detailed illustration is not essential for a proper understanding of the invention, should be illustrated in the drawing in the form of a graphical drawing symbol or a **labeled representation (e.g., a labeled rectangular box)**. In addition, tables and sequence listings that are included in the specification are, except for applications filed under 35 U.S.C. 371, not permitted to be included in the drawings.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as "amended." If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the

remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). If the changes are not accepted by the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Rejections - 35 USC § 112(a)

4. The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

5. Claim(s) 10-12 is/are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the enablement requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to enable one skilled in the art to which it pertains, or with which it is most nearly connected, to make and/or use the invention.

6. **Regarding claims 10 and 11:** The Applicant's disclosure does not enable one skilled in the art to apply the adjustment based on temperature and volume as set forth in claim 10. This is the only "calibration" listed in the specification therefore the lack of enablement applies to claim 11 as well. The specification does not show how one

obtains T_0 which is presumably the temperature inside the battery. The specification also does not specify how one knows the density of the gas. Para. 11 of the specification as published shows that the gas is a complicated mixture. Both of these values have to be known to apply the formula of claim 10 to adjust the measurement of volume determined by the measurement unit. The structures or processes necessary to obtain these values are not disclosed by the Applicant.

7. **Regarding claim 12:** Claim 12 recites that the processing device determines "parameters to be adjusted during formation of the battery." This same language is used several times throughout the specification however the Applicant does not give a single example of a parameter that can be adjusted based on this data. Claim 12 recites "parameters" plural, therefore if the specification does not enable a single parameter to be adjusted, it also does not enable multiple parameters to be adjusted during battery formation. Not only would the Applicant have to list what parameters could be adjusted during battery formation, but also would have to disclose the relationship between the volume of gas measured and the parameters to be adjusted in order to enable the limitation of determining "based on the analytical curve of the gas, parameters to be adjusted during formation of the battery."

Claim Rejections - 35 USC § 112(b)

8. The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

9. Claim(s) 2, 10, and 11 is/are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

10. **Regarding claims 2, 10, and 11:** It is improper to say, "calibrate the volume of gas." A volume of gas cannot be calibrated. Calibration is an adjustment made to measuring device to account for some aspect of the measurement. The claim should recite that data is adjusted based on the recited formula.

11. **Regarding claims 10 and 11:** Claims 10 and 11 rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being incomplete for omitting essential elements, such omission amounting to a gap between the elements. See MPEP § 2172.01. The omitted elements are: a temperature sensor to determine the initial temperature of the gas. How is T_0 determined? Furthermore, what sensor or programmed data is used to determine the density of the gas? Para. 11 of the specification as published lists many different gases that could be evolved from the battery formation. How is the density of this mixture known?

12. **Regarding claim 12:** This claim is rejected as indefinite for depending from an indefinite claim.

Claim Rejections - 35 USC § 102

13. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will

not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

14. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

15. Claim(s) 1, 3, and 4 is/are rejected under 35 U.S.C. 102(a1) as being anticipated by DU et al. (CN 216348930 U). Please note that a machine translation of DU has been included with this office action and is deemed sufficient. All references to text in DU are to the attached machine translation.

16. **Regarding claim 1:** DU discloses: A gas volume measurement apparatus (page 2 lines 41-43) for monitoring volume of gas produced during formation of a battery (page 1 lines 10-15), comprising: a delivery device (1, 2, 3, 4, 5, 6 in FIG. 1) and a measurement device (7, 9, 10 in FIG. 1); wherein: the delivery device connects the battery and the measurement device (page 3 lines 12-16) and is configured to remove hazardous substances from gas produced by the battery (by bubbling the gas through water as discussed on page 3 lines 21-30) and deliver the gas to the measurement device (7, 9); and the measurement device is configured to measure volume of the gas produced by the battery (page 2 lines 41-51).

17. **Regarding claim 3:** DU discloses: the delivery device comprises a gas-washing bottle (4 in FIG. 1), wherein a medium (water on page 4 lines 26-32) is accommodated in the gas-washing bottle, and the medium is used to dissolve the

hazardous substances in the gas (This is a statement of intended use. Since the gas bubbles up through the water as discussed on page 3 lines 28-30, it can dissolve hazardous substances in the gas.).

18. **Regarding claim 4:** DU discloses: the medium is water (page 4 lines 26-32).

Claim Rejections - 35 USC § 103

19. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

20. Claim(s) 5-9 is/are rejected under 35 U.S.C. 103 as being unpatentable over DU in view of LUO et al. (CN 203490042 U). Please note that a machine translation of LUO has been included with this office action and is deemed sufficient. All references to text in LUO are to the attached machine translation.

21. **Regarding claim 5-7:** DU does not disclose a buffer bottle.

22. LUO however does teach a buffer bottle (2 in FIG. 1), wherein no medium is accommodated in the buffer bottle (page 1 lines 42-50), and the buffer bottle (2) is disposed between the battery (on battery holder 9) and the gas-washing bottle (1). LUO also teaches rubber sealing plugs (11) with holes for delivery tubes 3, 4, and 5, thus meeting the limitations of claims 6 and 7.

23. One skilled in the art at the time the application was effectively filed would be motivated to use the buffer bottle of LUO in between the gas-washing bottle and the

battery of DU so that liquid will not get accidentally sucked into the battery, thus damaging the battery (page 1 lines 26-34 of LUO).

24. **Regarding claims 8 and 9:** DU and LUO teach most aspects of the instant invention. However, neither DU nor LUO explicitly teach the volume or size of the bottles. Nonetheless, the skilled artisan would know too that the size of the bottles would affect how accurate the volume measurement is, and how much room is needed for the apparatus

25. The specific claimed sizes, absent any criticality, is only considered to be the "optimum" sizes disclosed by DU and LUO that a person having ordinary skill in the art would have been able to determine using routine experimentation (see *In re Aller,* 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955)) based, among other things, on the desired accuracy of the volume measurement, manufacturing costs, etc. (see *In re Boesch,* 617 F.2d 272, 205 USPQ 215 (CCPA 1980)), and neither non-obvious nor unexpected results, i.e. results which are different in kind and not in degree from the results of the prior art, will be obtained as long as the sizes are used, as already suggested by DU and LUO.

26. Since the applicant has not established the criticality (see next paragraph) of the sizes stated and since these sizes are in common use in similar devices in the art, it would have been obvious to one of ordinary skill in the art at the time the invention was effectively filed to use these values in the device of DU as modified by LUO.

27. Please note that the specification contains no disclosure of either the critical nature of the claimed sizes or any unexpected results arising therefrom. Where patentability is said to be based upon particular chosen dimensions or upon another

variable recited in a claim, the applicant must show that the chosen dimensions are critical. *In re Woodruff,* 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990).

28. Claim(s) 2, 10, and 11 is/are rejected under 35 U.S.C. 103 as being unpatentable over DU in view of LEE et al. (KR 20210152885 A). Please note that a machine translation of LEE has been included with this office action and is deemed sufficient. All references to text in LEE are to the attached machine translation.

29. **Regarding claims 2, 10, and 11:** As best understood, DU does not teach a processing device.

30. LEE however does teach processing device (control unit on page 6 lines 41-52) for receiving values and making calculations (page 6 lines 41-52).

31. DU does not disclose applying the formula of claim 10, however the formula is simply the ideal gas law. Therefore the Examiner takes Official Notice that if one skilled in the art were measuring a volume of a gas under conditions where the pressure and temperature could change that the pressure and temperature must be taken into account to determine the volume of the gas because the volume of a gas has no meaning apart from temperature and pressure. The Examiner also takes Official Notice that it is also known to record data and plot the data on a graph, so that one can easily see many data points at once and recognize trends.

Conclusion

32. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. One or more references have been added to the Notice of References Cited for teaching an invention to measure a volume of gas produced by a battery.

33. Any inquiry concerning this communication or earlier communications from the examiner should be directed to NATHANIEL J KOLB whose telephone number is (571)270-7601. The examiner can normally be reached M-F 9-5 EST.

34. Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

35. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, JESSICA HAN can be reached at (571) 272-2078. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

36. Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/NATHANIEL J KOLB/
Examiner, Art Unit 2896